

$$y = \sqrt{x^3 - 2x}$$

$$\Rightarrow \sqrt{x^3 - 2x} = (x^3 - 2x)^{\frac{1}{2}}$$

$$\Rightarrow y' = \frac{1}{2} (x^3 - 2x)^{-\frac{1}{2}} (3x^2 - 2)$$

Simplify:

$$y' = \frac{(x^3 - 2x)^{-\frac{1}{2}}}{2} (3x^2 - 2)$$

$$y' = \frac{3x^2 - 2}{2\sqrt{x^3 - 2x}}$$

